

Data and AI in Sustainable Digital Transformation

From Brasília to the Plant Floor

Eduardo Sopalda · D&T Global Data Governance Lead · dsm-firmenich

1. The Architect

I am an architect. Not as a metaphor. I studied architecture, I drew buildings, and then I spent the last twenty years designing something less visible: data and AI governance.

I work inside a chemical company, like many of you, and I have got plenty of these things wrong myself. But I keep seeing the same problem: we design from above, not always for the person who has to live with the result.

2. Brasília

Let me take you to Brasília. From the air, it is extraordinary: monumental, precise, unforgettable. But at street level, it can feel very different. The scale is vast. The distances are large. A city can make perfect sense from a helicopter and feel completely different when you walk through it.

We build governance in a similar way. We have policies, controls, committees, operating models, and occasionally, a beautiful RACI that makes everyone feel civilisation has finally arrived.

For about ten minutes. Then someone asks who is actually accountable.

Then the framework reaches the plant.

I have been the person with the beautiful drawing. I have presented the framework, watched people nod, and then watched them carry on exactly as before. They were not being difficult. I had built them a plaza nobody wanted to stand in.

3. When ambition becomes evidence

That matters when we talk about sustainable digital transformation. We set ambitions, targets and roadmaps. We invest in systems, automation and AI. But eventually, ambition has to become evidence.

A carbon number looks like chemistry. It is a data product.

Behind one apparently simple number sit supplier declarations, energy data, production records, transport data, allocation rules and assumptions. Much of it comes from systems that were never designed to agree with one another.

So the challenge is not simply calculating the number. The challenge is being able to explain it. Where did it come from? Who owns it? What definition sits behind it? And who is prepared to stand behind it when someone asks a difficult question?

You cannot report what you cannot trace.

4. AI scales what is there

Then, inevitably, someone says: “Why do we not just use AI?”

I am optimistic about AI. But AI does not remove ambiguity. It can scale it.

If five people use the word “batch” to mean five different things, putting an AI agent on top does not make the organisation clearer. It just gives the confusion a faster processor.

Sometimes AI is the answer. Sometimes the answer is better data. Sometimes it is a better process. And occasionally it is simply agreeing on what a word means before we ask a machine to interpret it.

5. AI digital twins and smarter decisions

And take an AI digital twin.

It learns from the data of a real process: how equipment behaves, how materials respond, where energy is used, and where defects or waste occur.

Then it can help answer complex operational questions much faster. What if we change the temperature, the timing, or the material mix? Which settings could maintain quality while using less energy or producing less scrap?

Instead of relying on repeated physical trials, teams can use the AI digital twin to explore and compare possible outcomes before acting on the real process.

That can mean fewer wasted materials, less rework, and faster decisions about how to improve production.

This is one way AI can make sustainability practical: helping people find better ways to run the systems they already have, with less waste built in.

6. Close

So I want to leave you with one question.

What is one word in your organisation that means something different depending on which floor you are standing on?

Ask it at lunch. You may discover that the answer tells you more about your governance than another hundred pages of framework ever could.

Agents do not own risk. They do not carry accountability.

Humans do. Always.

Delivery cue: pause after “Then the framework reaches the plant,” “It is a data product,” and before the final line. Stop after “Humans do. Always.”